

Dirac Measure

Definition

Let $(X, \mathcal{A})$ be a measurable space and let $x \in X$. The Dirac measure at x , denoted by δ_x , is the measure defined by

$$\delta_x(A) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A, \end{cases} \quad A \in \mathcal{A}.$$

Interpretation

The Dirac measure places all its mass at the single point x . If $f : X \rightarrow \mathbb{R}$ is measurable, then

$$\int_X f d\delta_x = f(x),$$

whenever the integral is defined.

In Euclidean Space

On $\mathbb{R}^d$ with its Borel σ -algebra, δ_x is a probability measure concentrated at the point $x \in \mathbb{R}^d$.